

# Effects of chronic alkalinity stress on immune response in Nile tilapia (*Oreochromis niloticus*) infected with *Aeromonas hydrophila*

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## ABSTRACT

Under suitable environmental conditions, fish have a functional immune system and are therefore expected to have high disease resistance, growth rate and reproductive activity. To explore the effects of chronic alkalinity stress on the immune response in Nile tilapia, tilapia (291.1 ± 47.2 g) were exposed to 10 mmol/L and 30 mmol/L of sodium bicarbonate alkaline water (AW). At 31 d of alkalinity stress, the fish were infected by *Aeromonas hydrophila* and the cumulative mortality was calculated within 7 d. The immune-related indexes of peripheral blood cells, enzyme activities (Lys, Alp, C3 and IgM) and gene expression (*tlr5*, *il-1β* and *γ-ifn*) in the liver, spleen and gill of Nile tilapia were assessed at 1 and 31 d of alkalinity stress and 1 and 7 d post-infection. The results showed that: 1) the cumulative mortality of tilapia in 10 mmol/L AW was 26.7%, which was significantly lower than 66.7% in 30 mmol/L AW and 51.11% in freshwater (FW); 2) At 1 d post-infection, the respiratory burst and lymphocyte proliferation rates at 10 mmol/L AW were higher than at 30 mmol/L AW and FW; 3) Alp activity at 10 mmol/L AW was higher than at 30 mmol/L AW in the liver and spleen after infection; 4) At 31 d of alkalinity stress and 7 d post-infection, the *tlr5* mRNA expression level in the liver and spleen at 10 mmol/L AW was higher than at 30 mmol/L AW and FW. At 7 d post-infection, the *γ-ifn* mRNA expression level in the gill of tilapia in 10 mmol/L AW was higher than in 30 mmol/L AW and FW. The *il-1β* mRNA expression level at 30 mmol/L AW was significantly higher than at 10 mmol/L AW at most time points in the spleen and gill; 5) At 7 d post-infection, the activities or expression level of Lys, Alp, C3, IgM, *tlr5* and *γ-ifn* in the spleen of tilapia in 30 mmol/L AW were lower than in 10 mmol/L AW and FW. The above results showed that long-term feeding of Nile tilapia in 10 mmol/L carbonate AW can promote immune function and reduce mortality from bacterial infection.

## 1. Introduction

The effects of environmental stress on the immune response have been studied for many decades (Elenkov and Chrousos, 2006; Simi et al., 2017). Inappropriate environmental conditions may cause acute or chronic stress to fish and suppress specific defence systems. On the contrary, under appropriate environmental conditions, fish have a functional immune system and thus are expected to have high disease resistance, growth rate and reproductive activity. For example, it is considered that fish possess an appropriate thermal range for immunomodulation (Domínguez et al., 2004; Wang et al., 2020). When Nile tilapia were reared at 21 °C, the immune response and vaccine efficacy

were significantly inhibited at 28 d post-infection (Wang et al., 2020). Rearing Nile tilapia at 33 °C also led to a decrease in IgM concentration at 4 weeks (Domínguez et al., 2004). Contrarily, for Nile tilapia reared at 18.4, 23 and 28 °C, the IgM concentration increased with increased water temperature after 2 weeks (Domínguez et al., 2004). Another example of improved immunity is the significant interrelated increases of plasma IgM in Atlantic cod (*Gadus morhua* L.) reared at 1, 7 and 14 °C for 12 months (Magnadóttir et al., 1999). As with the temperature range, fish also have an appropriate immunomodulatory salinity range. Salinity at ≥20 ppt has a negative effect on the antibodies of Nile tilapia, and finally affected the efficacy of a vaccine at 28 d post-infection (Wang et al., 2018). Contrarily, the plasma IgM level of Nile tilapia increased

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significantly with salinity at 12 ppt after 4 weeks (Domínguez et al., 2004). In addition to the intensity of stress, another factor to consider is the duration of stress, i.e., acute or chronic stress. The acute stress activation of the innate immune system may be used as an adjuvant to enhance the immune response to antigen challenge. Therefore, it is generally believed that acute stress is beneficial to mammals and bony fish. The effect of chronic stress is a little more complicated (Edwards et al., 2007; Fast et al., 2008; Nardocci et al., 2014). In general, it seems that each fish species has a specific tolerance range to different environmental conditions (stress intensity, stress duration). Therefore, to promote aquaculture, it is very important to evaluate the relationship between changes in environmental conditions and the impact on the fish immune system.

The alkalinity of carbonate is a critical abiotic factor in aquaculture farming affecting the normal growth and development of fish (Yao et al., 2012). The alkalinity of natural water is mainly composed of carbonate and bicarbonate. Appropriate alkalinity can stabilize the pH value of water, improve the buffer capacity and maintain the stability of the pond environment. High carbonate alkalinity has strong toxic effects on fish, such as gill tissue damage, hypocapnia, respiratory alkalosis and ammonia poisoning (Wilkie and Wood, 1996). Nile tilapia (*Oreochromis niloticus*), known as euryhaline fish, can tolerate a wider range of salt and alkalinity compared with ordinary freshwater fish (Zhao et al., 2015). Transcriptome analysis showed that several lipid metabolism (involving signal transduction), detoxification and immune-related genes displayed significant changes upon acute high carbonate alkalinity stress (40–60 mmol/L) in tilapia (Zhao et al., 2020). When Nile tilapia were exposed to saline-AW (10 ppt, 14–42 mmol/L), they exhibited nutrient adaptations in digestion and absorption (Song et al., 2021). Alkalinity stress also resulted in a significant decrease in the level of *miR-21*, which could affect the balance of ion regulation, reactive oxygen species (ROS) production and antioxidant enzyme activity in tilapia (Zhao et al., 2016). However, there is little information about the influence of alkalinity on immunomodulation in Nile tilapia.

Therefore, in this study, Nile tilapia was exposed to 10 mmol/L and 30 mmol/L of sodium bicarbonate AW for 31 days, and then the fish were infected with *Aeromonas hydrophila* which is a common pathogen that can cause tissue damage, necrosis, hydrops and other symptoms in fish (Liu et al., 2021). The cumulative mortality was calculated within 7 days. The activities of leukocytes in peripheral blood, immune-related enzymes activities (lysozyme, alkaline phosphatase, complement component 3 and immunoglobulin M) and gene expression (Toll-like receptor 5, interleukin 1 $\beta$  and  $\gamma$ -interferon) in the liver, spleen and gill of tilapia were measured at 1 and 31 d post alkalinity stress and 1 and 7 d post-infection to explore the effect of long-term alkalinity stress on immunomodulation in tilapia and to provide evidence for healthy farming and disease prevention and control.

## 2. Materials and methods

### 2.1. Experimental fish and bacteria strains

This study was approved by the animal ethics committee of Shanghai Ocean University. The experiment was conducted at the tilapia germplasm resources station of Shanghai Ocean University from May 1, 2021. The fish (291.1  $\pm$  47.2 g) were the offspring of the GIFT strain of Nile tilapia (*Oreochromis niloticus*). Individuals with a strong physique were selected for alkalinity stress. The water [NaHCO<sub>3</sub> (analytical pure)] was configured in advance and was stable and oxygen-rich.

The *Aeromonas hydrophila* used for infection was from the Key Laboratory of Exploration and Utilization of Aquatic Genetic Resources, Ministry of Education, Shanghai Ocean University.

### 2.2. Determination of median lethal concentration

The *Aeromonas hydrophila* glycerol bacteria frozen at  $-80^{\circ}\text{C}$  were

cultured into Luria-Agar (LA) medium at  $28^{\circ}\text{C}$  for 24 h, then single colonies were selected and inoculated into Luria-Bertani (LB) medium at  $28^{\circ}\text{C}$  with rotational speed of shaker 2000 rpm/min overnight for facultative anaerobic culture. After resuspension with 0.85% sterile normal saline (pH 7.2), the absorbance of the bacterial solution at OD600 nm was measured by an ultraviolet spectrophotometer (Quawell Technology Inc. San Jose, CA, USA), and the concentration of the bacterium solution was determined by a quantitative standard curve. The bacteria count was given as CFU/mL of bacterial solution. The concentration of original bacterial solution was  $3.0 \times 10^8$  CFU/mL. To determine the median lethal concentration within 7 days (LC50-7d) of bacteria for tilapia in freshwater (FW), fifty fish were divided randomly into one negative control group (peritoneally injected, 0.2 mL normal saline) and four experimental groups (peritoneally injected, 0.2 mL original bacterial solution and 1/2, 1/4, 1/8 diluted bacterial solution). The water temperature was maintained at  $25 (\pm 1)^{\circ}\text{C}$ . At 7 d post-infection, the cumulative mortality at different concentrations of *Aeromonas hydrophila* was determined, and the calculated LC50-7d was  $8.4 \times 10^7$  CFU/mL (Colony-Forming Units, CFU).

### 2.3. Experimental design and sample collection

Previous experiments showed that the 96 h half lethal alkalinity of Nile tilapia with an average body mass of 21.56 ( $\pm$  0.27) g was about 100 mmol/L (Regression formula:  $y = 0.329x - 2.445$ ), and the 96 h half lethal alkalinity of Nile tilapia with an average body mass of 5.92 ( $\pm$  0.03) g was about 70 mmol/L (Regression formula:  $y = 0.200x - 0.750$ ). Thus, Nile tilapia can survive for a long time exposed to 10 mmol/L and 30 mmol/L of sodium bicarbonate AW. Before the treatment, all the fish were kept in FW and fasted for 24 h (initial group). Overall, fish were divided into three experimental groups by random: some fish were maintained in FW and the other fish were transferred directly from freshwater (FW) to 10 mmol/L and 30 mmol/L AW. Three repetitions were set in different experimental groups, with 15 fish in each repetition. All fish were reared in FW and AW for 31 days and fed twice daily. At 31 d post alkalinity stress, fish were infected by 0.2 mL *Aeromonas hydrophila* at a concentration that led to half the deaths of tilapia in freshwater (LC50-7d:  $8.4 \times 10^7$  CFU/mL). After a peritoneal injection, they continued to be reared in the FW and AW group respectively. The cumulative mortality within 7 days was calculated (no feed after infection). Samples (blood, liver, spleen and gill) were collected from the initial group, after 1 and 31 d of alkalinity stress and 1 and 7 d post-infection. Blood was collected with anticoagulant for the isolation of leukocytes. The isolated liver, spleen and gill were used to measure enzyme activity and gene expression. During the experiment, the water was replaced every 24 h and the dissolved oxygen was about 6 mg/L and water temperature was about  $23\text{--}26^{\circ}\text{C}$ .

### 2.4. Respiratory burst analysis

The content of ROS in monocytes was measured with the fluorescent labelling probe, DCFH-DA (2',7'-Dichlorodihydrofluorescein diacetate, Sigma, USA). Mix blood cell suspensions of 200  $\mu\text{L}$  and DCFH-DA (10  $\mu\text{M}$ ) were stored at  $37^{\circ}\text{C}$  in a dark location for 30 min. After filtering through a 200-mesh screen, the suspension was tested by flow cytometry as soon as possible. In total, 1000 cells were obtained from each sample. The first fluorescence channel (FL1) was used to obtain the intracellular DCF (2',7'-Dichlorofluorescein) fluorescence data. Finally, we established a single parameter histogram with DCF fluorescence (FL1) as the abscissa and cell count as the ordinate using CellQuest software to analyze the average fluorescence of DCF (Zhang et al., 2015).

### 2.5. Cell proliferation analysis

Peripheral blood (0.2 mL) was added to phosphate-buffered saline (PBS, pH = 7.4, 1 mL) for washing and the supernatant was discarded.

Then a trypsin-EDTA solution containing 0.25% trypsin was added. When cells become round and partially suspended, PBS was added to terminate the digestion. The cell suspension blown by a pipette gun was centrifuged at 1500 rpm for 5 min to collect cells (Tang et al., 2013). Next, 3 mL of PBS were added to resuspend the cells and collect precipitates by centrifugation at intervals following the same protocol. Then, 75% ethanol precooled at  $-20^{\circ}\text{C}$  was added slowly and left overnight at  $4^{\circ}\text{C}$ . The cells were washed in PBS again and then 2  $\mu\text{L}$  RNase A (1 mg/mL) was added and the suspension was left in a water bath at  $37^{\circ}\text{C}$  for 40 min. Propidium iodide (100  $\mu\text{L}$ , 100  $\mu\text{g/mL}$ ) was added in the dark for 20 min. Finally, the cells were tested by flow cytometry (FCM, FC500, USA) with an excitation and emission wavelength at 488 nm and 585 nm, respectively (Yang et al., 2022).

## 2.6. Enzyme activity analysis

Tissue (0.3 g) was frozen at  $-80^{\circ}\text{C}$  and mixed with normal saline (precooled, sterilized) to a mass volume ratio of 1:9 for tissue homogenate, then the supernatant was taken from the sample for testing after centrifugation. Immune-related enzymes activities (lysozyme, alkaline phosphatase, complement component 3 and immunoglobulin M) were measured using an ELISA Kit (Enzyme-linked Biotechnology Co., Ltd., Shanghai, China).

## 2.7. RNA extraction, cDNA synthesis and qPCR

Total RNA was extracted from Nile tilapia tissues according to the instructions provided with the Trizol reagent (Invitrogen, Tiangen Biotech Co., Ltd., Beijing, China) and then RNA concentration and integrity were determined (Zhou et al., 2021). Total RNA was used to synthesize cDNA by reverse transcription using Prime Script RT (Accurate Biotechnology Co., Ltd., Hunan, China). The cDNA was stored at  $-20^{\circ}\text{C}$ . Real-time Quantitative PCR (RT-qPCR) was performed to detect mRNA expression levels using SYBR Green Premix Pro Taq HS qPCR Kit (Accurate Biotechnology Co., Ltd., Hunan, China) as instructed. The primer pairs of *tlr5* (Pang, 2016), *il-1 $\beta$*  (Kayansamruaj et al., 2014),  *$\gamma$ -ifn* (Luo et al., 2015) and  *$\beta$ -actin* as housekeeping gene (Song et al., 2021) were synthesized as described previously (Table 1). Finally, the  $2^{-\Delta\Delta\text{Ct}}$  method was used to determine the relative expression of each gene (Livak and Schmittgen, 2002).

## 2.8. Data analysis

The results were expressed as the mean  $\pm$  standard deviation (SD). GraphPad Prism 8.0 was used to plot histograms. The two-way ANOVA, followed by Fisher's post hoc test were performed with SPSS 25.0 to evaluate the significant differences within and between groups for the results of respiratory burst, cell proliferation, enzyme activity and gene expression. The one-way ANOVA, followed by Fisher's post hoc test were conducted to determine significant differences for the results of cumulative mortality. The results were considered as significant at  $P < 0.05$ .

**Table 1**  
Primer used for RT-qPCR.

| Gene                            | Primer sequence (5' to 3')                        | Primer efficiency(%) | Amplicon size (bp) | Annealing temperature ( $^{\circ}\text{C}$ ) |
|---------------------------------|---------------------------------------------------|----------------------|--------------------|----------------------------------------------|
| <i>tlr5</i>                     | F/CATTACGCGTCTCCCTAACT<br>R/CGACATGGATACTGTTCATGG | 96.9                 | 168                | 54.6                                         |
| <i>il-1<math>\beta</math></i>   | F/GCTGTGGAGAAGAACCAAGC<br>R/TGCTGAGCACAGAAATCCAG  | 98.1                 | 371                | 59.5                                         |
| <i><math>\gamma</math>-ifn</i>  | F/CAGCAGAGATGAACCTTGA<br>R/CACTAGGAAATACGGGTTT    | 97.3                 | 128                | 60.0                                         |
| <i><math>\beta</math>-actin</i> | F/TGGTGGGTATGGGTCAGAAAG<br>R/CTGTGGCTTTGGGGTTCA   | 99.3                 | 217                | 58.2                                         |

The expression levels of *18S rRNA*, *GAPDH* and  *$\beta$ -actin* were determined as candidate internal reference genes, and their stability was evaluated by geNorm and Norm Finder software. Finally,  *$\beta$ -actin* was selected as an internal reference gene.

## 3. Results

### 3.1. Cumulative mortality of Nile tilapia infected with *Aeromonas hydrophila*

No deaths were recorded before injection. At 17 h post-infection, fish began to die. Sick fish had prominent symptoms such as decreased activity, white fins, congested gills, etc. At 7 d post-infection, the cumulative mortality at FW, 10 mmol/L AW and 30 mmol/L AW was 51.11%, 26.7% and 66.7%, respectively.

### 3.2. Peripheral blood leukocyte activity

Two-way ANOVA showed that there was a statistically significant interaction between the effects of alkalinity and sampling point on respiratory burst, monocyte and lymphocyte proliferation respectively ( $P < 0.05$ ). All the effect sizes (partial eta-squared) of alkalinity were above 0.14.

#### 3.2.1. Monocyte respiratory burst

At 1 d post alkalinity stress, the respiratory burst activity of monocytes at 10 mmol/L AW increased sharply, but that of 30 mmol/L AW remained predominantly unchanged. At 31 d post alkalinity stress, the respiratory burst activity of 10 mmol/L AW reached a peak and was significantly higher than that of 30 mmol/L AW ( $P < 0.05$ ). The respiratory burst activity of monocytes at 10 mmol/L AW decreased at 1 d post-infection, but it was still significantly higher than that of FW and 30 mmol/L AW ( $P < 0.05$ ). During the experiment, the respiratory burst activity of 30 mmol/L AW was always close to the initial level (Fig. 1).

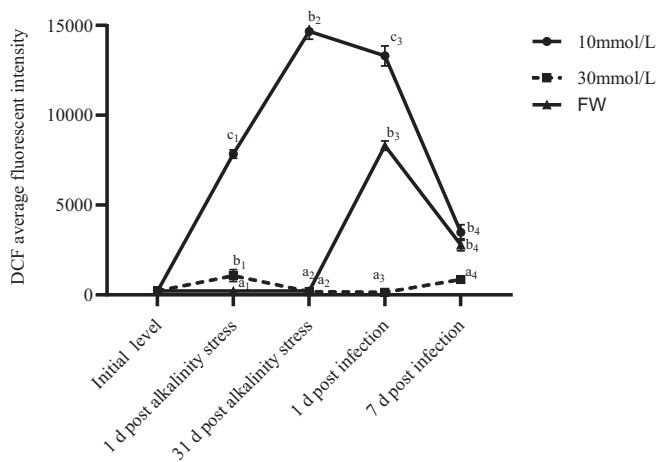
#### 3.2.2. Monocyte and lymphocyte proliferation

The proliferation of monocytes with different alkalinity had a similar trend during the experiment. They increased at 1 d of alkalinity stress and decreased at 31 d of alkalinity stress, and increased at 1 d post-infection and decreased at 7 d post-infection. The proliferation rate of monocytes in the 10 mmol/L AW group was close to that in the 30 mmol/L AW group.

At 1 d of alkalinity stress, the lymphocyte proliferation rate of different alkalinity groups all increased. The lymphocyte proliferation rate of fish in 10 mmol/L AW increased but that of fish in 30 mmol/L AW decreased after infection. At 1 d of alkalinity stress and 1 d post-infection, the lymphocyte proliferation rate of 10 mmol/L AW was significantly higher than that of FW and 30 mmol/L AW ( $P < 0.05$ ). The other sampling points were predominantly the same (Fig. 2).

### 3.3. Activities/levels of immune-related enzymes

Two-way ANOVA showed that there was a statistically significant interaction between the effects of alkalinity and sampling point on activity of each immune-related enzyme in a specific tissue ( $P < 0.05$ ). Effect sizes (partial eta-squared) of alkalinity were as follows: 0.245 (Lys in liver), 0.922 (Lys in spleen), 0.491 (Lys in gill), 0.053 (Alp in liver),

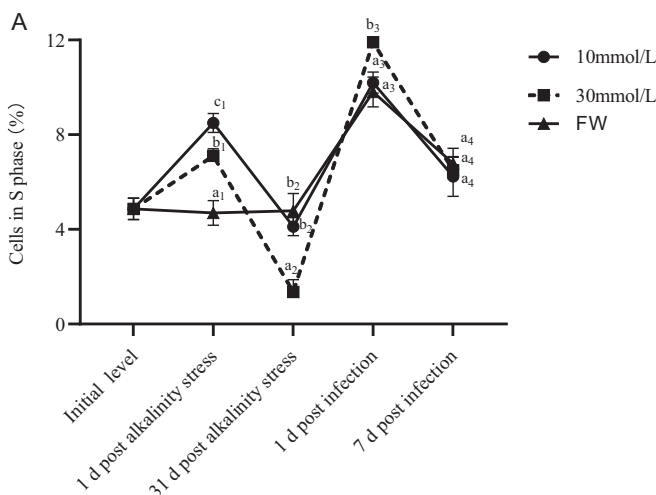


**Fig. 1.** Respiratory burst activity of peripheral blood monocytes in the 10 mmol/L AW, 30 mmol/L AW and FW groups, compared to that of FW group. The values are given as mean  $\pm$  SD for separate fish per group ( $n = 3$ ). Different subscript letters indicate significant differences ( $P < 0.05$ ).  $a_1/b_1/c_1$  denotes the difference among different alkalinity at 1 d of alkalinity stress,  $a_2/b_2/c_2$  denotes the difference among different alkalinity at 31 d post alkalinity stress,  $a_3/b_3/c_3$  denotes the difference among different alkalinity at 1 d post-infection,  $a_4/b_4/c_4$  denotes the difference among different alkalinity at 7 d post-infection.

0.514 (Alp in spleen), 0.128 (Alp in gill), 0.550 (C3 in liver), 0.342 (C3 in spleen), 0.483 (C3 in gill), 0.591 (IgM in liver), 0.519 (IgM in spleen) and 0.066 (IgM in gill).

### 3.3.1. Lysozyme (Lys)

At 1 d of alkalinity stress, Lys activity in different alkalinity groups decreased. At 31 d of alkalinity stress, Lys activity of 30 mmol/L AW was significantly higher in the liver, but lower in the gill and spleen than that of FW ( $P < 0.05$ ). At 1 d post-infection, Lys activity in alkalinity groups was lower in the spleen than in the FW group ( $P < 0.05$ ). There was no significant difference between different alkalinity groups in different tissues. At 7 d post-infection, the Lys activity of the fish in 30 mmol/L AW was lower in the spleen than that of the fish in FW and 10 mmol/L AW, whereas it is the opposite in the liver ( $P < 0.05$ ) (Fig. 3).



**Fig. 2.** Proliferation changes of monocytes (A) and lymphocytes (B) in the peripheral blood of 10 mmol/L AW and 30 mmol/L AW groups, compared to that of FW group. The values are given as mean  $\pm$  SD for separate fish per group ( $n = 3$ ). Different subscript letters indicate significant differences ( $P < 0.05$ ).

### 3.3.2. Alkaline phosphatase (alp)

At 1 d of alkalinity stress, Alp activity changed and was higher in the 30 mmol/L AW group than in the 10 mmol/L AW group in the liver. At 31 d of alkalinity stress, Alp activity increased to a higher level than in the initial group in the liver but decreased in the spleen, and tended to return to the initial level in the gill. At 1 d post-infection, Alp activity in the FW and 10 mmol/L AW group was significantly higher in the liver than that in the 30 mmol/L AW group ( $P < 0.05$ ). At 7 d post-infection, Alp activity in the FW and 10 mmol/L AW group was significantly higher in the spleen than that in the 30 mmol/L AW group ( $P < 0.05$ ) (Fig. 4).

### 3.3.3. Complement component 3 (C3)

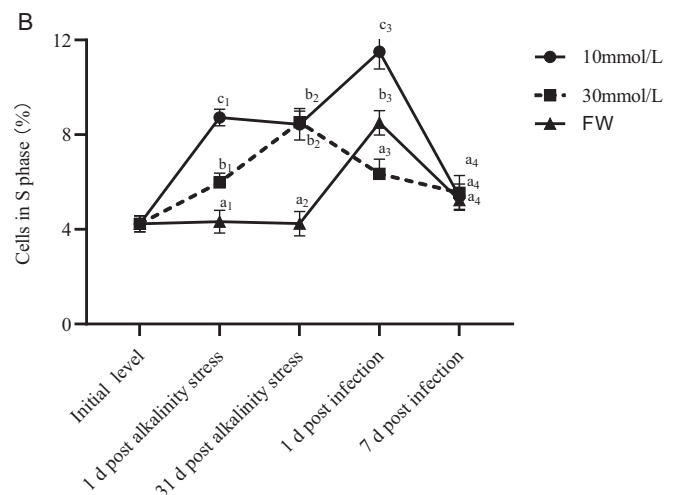
At 1 d of alkalinity stress, the C3 levels of different alkalinity groups decreased in the liver, but increased in the spleen, and increased significantly in the 30 mmol/L AW group in the gill. At 31 d of alkalinity stress, the level of C3 in each tissue was lower than or close to the initial level, and there was no significant difference between different alkalinity groups ( $P > 0.05$ ). At 1 d post-infection, the level of C3 in different alkalinity groups increased in the liver. At 7 d post-infection, the level of C3 in the 30 mmol/L AW group was significantly lower in the spleen but higher in the gill than that in the 10 mmol/L AW group ( $P < 0.05$ ) (Fig. 5).

### 3.3.4. Immunoglobulin M (IgM)

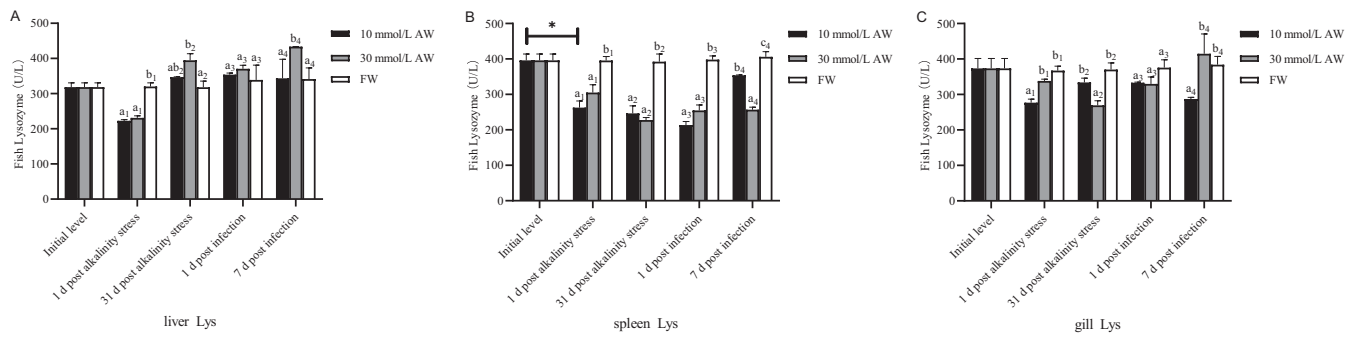
At 1 d of alkalinity stress, the IgM levels of different alkalinity groups in the liver and spleen decreased, and IgM rose to 1.7 times that in the initial group in the gill of the 30 mmol/L AW group. At 31 d of alkalinity stress, IgM activity in the 10 and 30 mmol/L AW group was significantly lower in the spleen than that in the FW group ( $P < 0.05$ ). At 1 d post-infection, the IgM level in the gill of the FW and 10 mmol/L AW group was higher than that of the 30 mmol/L AW group. At 7 d post-infection, the IgM levels in the liver and gill of different alkalinity groups were close to the FW group, and the IgM in the spleen of the 30 mmol/L AW group was significantly lower than that of FW and 10 mmol/L AW group ( $P < 0.05$ ), and dropped to 0.15 times that of the initial group (Fig. 6).

### 3.4. Immune-related gene expression

Two-way ANOVA showed that there was a statistically significant interaction between the effects of alkalinity and sampling point on expression of each immune-related gene in a specific tissue ( $P < 0.05$ ). Effect sizes (partial eta-squared) of alkalinity were as follows: 0.737 (*thr5* in liver), 0.960 (*thr5* in spleen), 0.861 (*thr5* in gill), 0.063 (*il-1 $\beta$*  in liver),

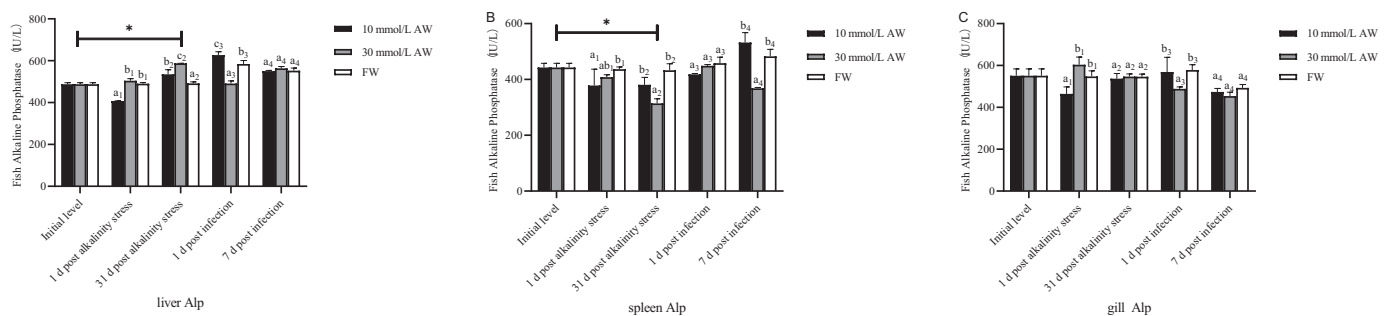




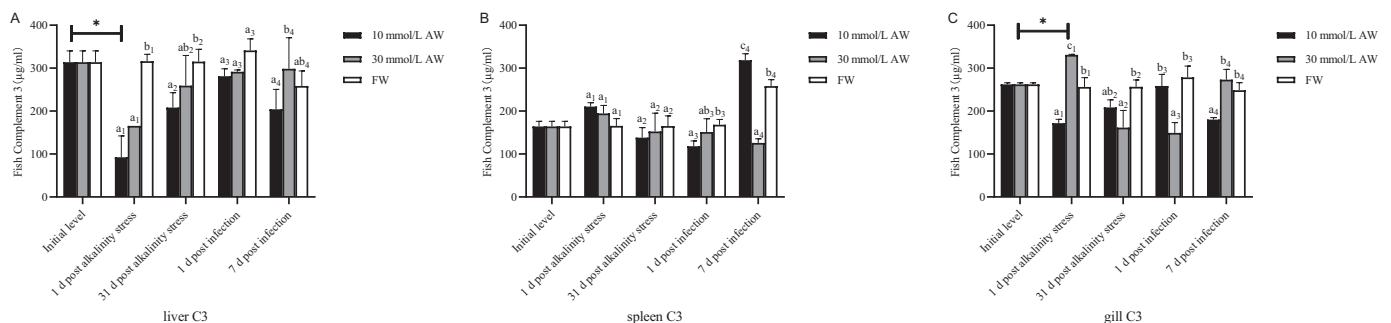


**Fig. 3.** Lys activities in liver (A), spleen (B) and gill (C) of the 10 mmol/L AW, 30 mmol/L AW and FW groups. The values are given as mean  $\pm$  SD for separate fish per group (n = 3). Different subscript letters indicate.

Significant differences ( $P < 0.05$ ).  $a_1/b_1/c_1$  denotes the difference among different alkalinity at 1 d of alkalinity stress,  $a_2/b_2/c_2$  denotes the difference among different alkalinity at 31 d post alkalinity stress,  $a_3/b_3/c_3$  denotes the difference among different alkalinity at 1 d post-infection,  $a_4/b_4/c_4$  denotes the difference among different alkalinity at 7 d post-infection. \* indicate significant differences ( $P < 0.05$ ) among different sampling point at same alkalinity.



**Fig. 4.** Alp activities in the liver (A), spleen (B) and gill (C) of the 10 mmol/L AW, 30 mmol/L AW and FW groups. The values are given as mean  $\pm$  SD for separate fish per group (n = 3).



**Fig. 5.** C3 activities in the liver (A), spleen (B) and gill (C) of the 10 mmol/L AW, 30 mmol/L AW and FW groups. The values are given as mean  $\pm$  SD for separate fish per group (n = 3).

0.978 (*il-1 $\beta$*  in spleen), 0.996 (*il-1 $\beta$*  in gill), 0.339 ( *$\gamma$ -ifn* in liver), 0.033 ( *$\gamma$ -ifn* in spleen) and 0.949 ( *$\gamma$ -ifn* in gill).

### 3.4.1. Toll-like receptor 5 (*tlr5*)

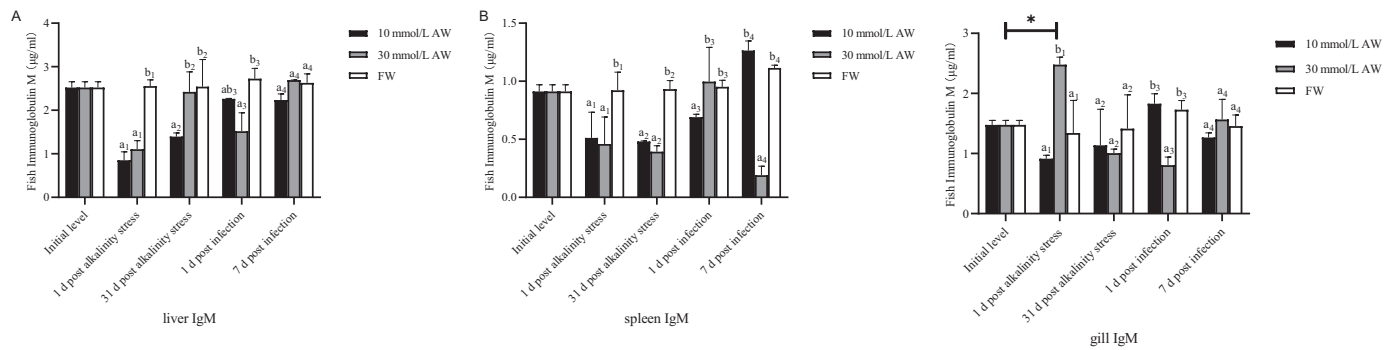
At 1 d of alkalinity stress, the *tlr5* mRNA expression level increased in the spleen and gill of the different alkalinity groups. The *tlr5* mRNA expression level in the liver of the 30 mmol/L AW group decreased significantly and was 0.1 times that of the initial group. At 31 d of alkalinity stress, the *tlr5* mRNA expression level in the liver and spleen of the 10 mmol/L AW group was higher than in the FW and 30 mmol/L AW group ( $P < 0.05$ ). However, the *tlr5* mRNA expression level in the gill at 10 mmol/L AW was significantly lower than at 30 mmol/L AW. At 7 d post-infection, the *tlr5* mRNA expression level in the liver and spleen of the 10 mmol/L AW group increased and was higher than that of the FW and 30 mmol/L AW group ( $P < 0.05$ ) (Fig. 7).

### 3.4.2. Interleukin 1 $\beta$ (*il-1 $\beta$* )

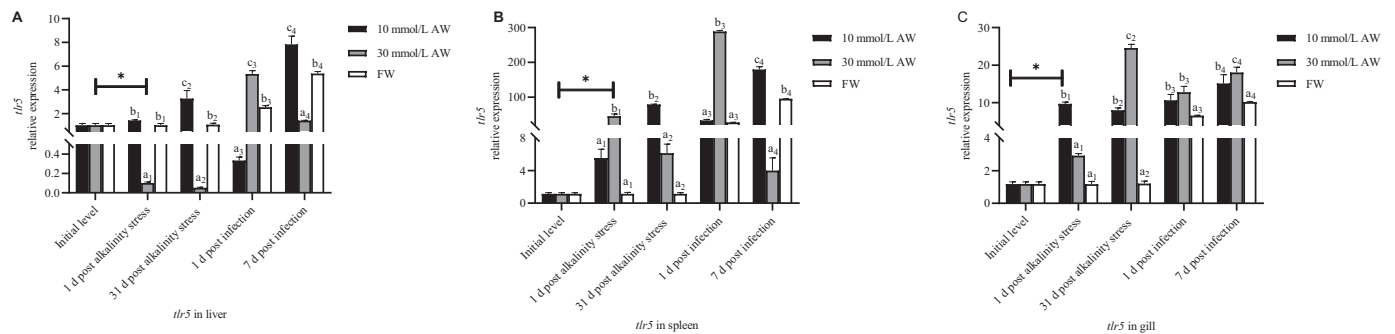
At 1 d of alkalinity stress, the *il-1 $\beta$*  mRNA expression level changed in the liver. The *il-1 $\beta$*  mRNA expression level in the spleen and gill of the 10 mmol/L AW group was close to the initial level, but that of the 30 mmol/L AW increased to a very high level. At 31 d post alkalinity stress, the *il-1 $\beta$*  mRNA expression in the spleen and gill of the 30 mmol/L AW group was significantly higher than that of the FW and 10 mmol/L AW ( $P < 0.05$ ), whereas it is the opposite in the liver. At 1 and 7 d post-infection, the *il-1 $\beta$*  mRNA expression level in the spleen and gill of the 30 mmol/L AW group was significantly higher than that of the FW and 10 mmol/L AW group ( $P < 0.05$ ) (Fig. 8).

### 3.5. $\gamma$ -gamma interferon ( *$\gamma$ -ifn*)

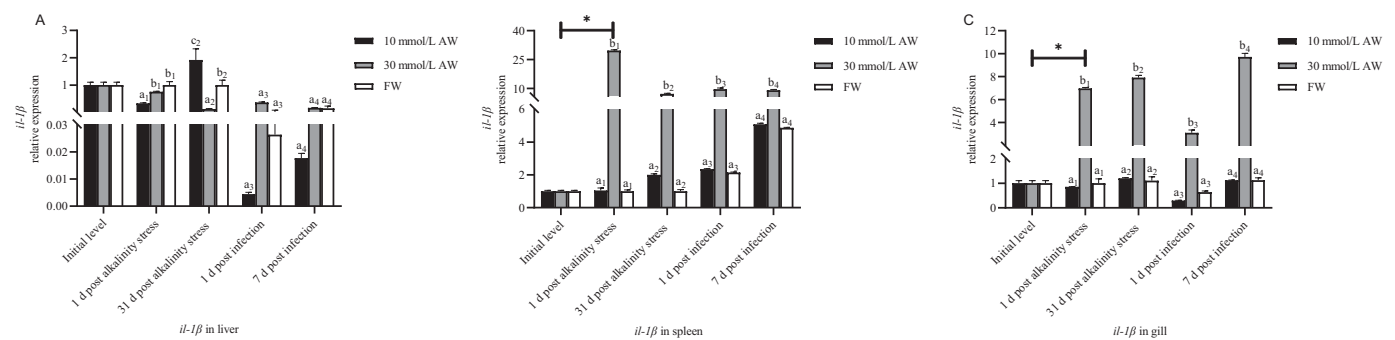
At 1 d of alkalinity stress, the  *$\gamma$ -ifn* mRNA expression level in different



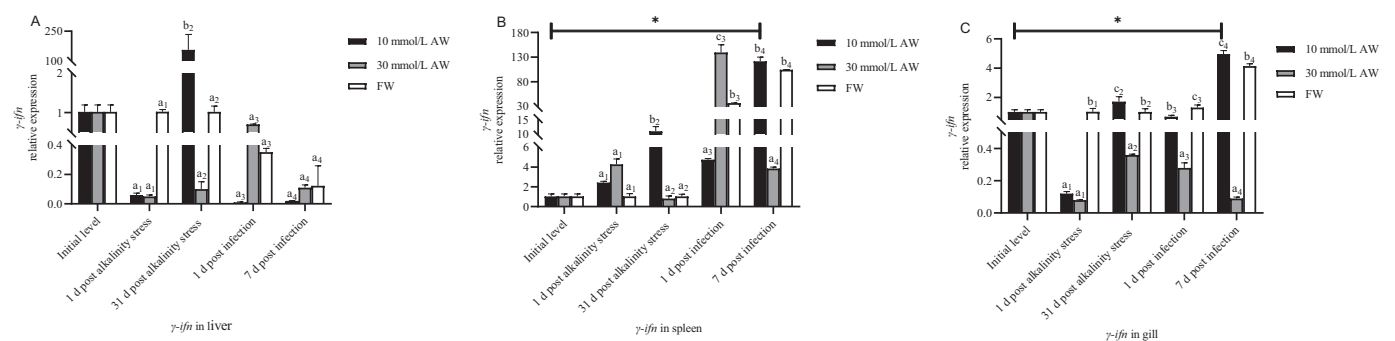
**Fig. 6.** IgM activities in the liver (A), spleen (B) and gill (C) of the 10 mmol/L AW, 30 mmol/L AW and FW groups. The values are given as mean  $\pm$  SD for separate fish per group ( $n = 3$ ).



**Fig. 7.** Relative expression of *thr5* in the liver (A), spleen (B) and gill (C) of the 10 mmol/L AW, 30 mmol/L AW and FW groups. The values are given as mean  $\pm$  SD for separate fish per group ( $n = 3$ ).



**Fig. 8.** Relative expression of *il-1β* in the liver (A), spleen (B) and gill (C) of the 10 mmol/L AW, 30 mmol/L AW and FW groups. The values are given as mean  $\pm$  SD for separate fish per group ( $n = 3$ ).



**Fig. 9.** Relative expression of  $\gamma$ -*ifn* in the liver (A), spleen (B) and gill (C) of the 10 mmol/L AW and 30 mmol/L AW and FW groups. The values are given as mean  $\pm$  SD for separate fish per group ( $n = 3$ ).

alkalinity groups decreased in the liver and gill but increased in the spleen. At 31 d of alkalinity stress, the  $\gamma$ -*ifn* mRNA expression level of different alkalinity groups showed an upward trend in the liver and gill. The  $\gamma$ -*ifn* mRNA expression level in the spleen of the 10 mmol/L AW group increased but that of the 30 mmol/L AW group decreased. In all tissues, the  $\gamma$ -*ifn* mRNA expression level of the fish in 10 mmol/L AW was higher than that in 30 mmol/L AW which was lower than or close to the initial level. At 7 d post-infection, the  $\gamma$ -*ifn* mRNA expression level in the liver of the 10 mmol/L AW group stayed incredibly low but increased in the spleen and gill and was higher than in the 30 mmol/L AW group and initial group ( $P < 0.05$ ) (Fig. 9).

#### 4. Discussion

Chronic carbonate alkalinity stress of 15 mmol/L could significantly reduce the activities of complement and IgM in the intestine and hepatopancreas of *Cyprinus carpio* Songpu (Fan et al., 2021). Upon acute high alkalinity stress, the leukocyte phagocytic activity and plasma Lys activity of *Carassius auratus gibelio* decreased significantly (Li et al., 2011). In this study, the cumulative mortality of tilapia in 10 mmol/L AW (26.7%) was significantly lower than that in FW (51.1%) and 30 mmol/L AW (66.7%) at 7 d post-infection. After infection, the immune responses of tilapia in 10 mmol/L AW and 30 mmol/L AW were different. Respiratory burst activity, C3 and *thr5* activities in the spleen of the 10 mmol/L AW group were significantly higher than those of the 30 mmol/L AW and FW groups at 7 d post-infection. Alp, IgM and  $\gamma$ -*ifn* activities in the spleen of the 10 mmol/L AW group were close to FW and higher than those of the 30 mmol/L AW groups at 7 d post-infection. These results strongly suggest that long-term low alkalinity (10 mmol/L) can improve the immune function of tilapia against *Aeromonas hydrophila*, ultimately reducing the mortality of infected tilapia. Although long-term high alkalinity (30 mmol/L) can increase some immunological indexes, it did not increase resistance to pathogens. Furthermore, 10 mmol/L alkalinity is a safe farming concentration for tilapia and can be used in production to reduce the impact of some bacterial diseases.

Leukocytes in fish peripheral blood are important components of cellular and humoral immunity. Detecting the activity of leukocytes in peripheral blood is an important index to evaluate immune function (Wlasow and Dabrowska, 1989). Monocytes are transformed into phagocytes during stress to enhance phagocytosis, thus phagocytizing bacteria invade the body (Akira et al., 2006). Lymphocytes can be induced to produce corresponding antibodies and various immune factors after antigen stimulation (Akira et al., 2006). Ye et al. (2011) found that in Nile tilapia ( $22.0 \pm 1.0$  g) infected by *Aeromonas hydrophila* ( $1 \times 10^6$  CFU/fish), the number of leukocytes, NBT positive cells, and serum lysozyme activity increased first and then decreased within 3 d, and returned to the normal level at a later stage. In this study, after the exposure to alkalinity stress, the respiratory burst activity of monocytes and the proliferation rate of monocytes and lymphocytes increased in different alkalinity groups. After infection, the respiratory burst activity of monocytes and lymphocyte proliferation rate of the 10 mmol/L AW group were significantly higher than those of the 30 mmol/L AW and FW groups, indicating that carbonate alkalinity stress influenced cellular immunity. Long-term low carbonate alkalinity stress (10 mmol/L) can improve immune performance in tilapia by improving leukocyte activity.

Lys, Alp, C3 and IgM are important immune indexes in fish. Environmental stress and diseases can lead to their corresponding changes. Lys and Alp mainly function in immune defence by destroying microbial cell walls (Ayiku et al., 2020), regulating calcium and phosphorus metabolism and participating in protein synthesis (Campbell et al., 2010). C3 binding to pathogens through recognition proteins promotes macrophage phagocytosis and further promotes the release of cytokines and the production of antibodies (Gallani et al., 2019). IgM is the main immunoglobulin of fish, which can recognize and neutralize invading

viruses and bacteria in vitro (Cerezuela et al., 2012). The concentration of the C3/IgM in the spleen of grass carp (*Ctenopharyngodon idella*) increased significantly 14 days post-infection with *Aeromonas hydrophila* (Liu et al., 2018). In this study, Alp activity showed an upward trend in liver tissues during alkalinity stress. The Alp activity of the 10 mmol/L AW group was higher than that of the 30 mmol/L AW group in the liver and spleen after infection, which reflected the important role of Alp in disease resistance. Compared with this, other enzyme activities lacked unified regularity among different alkalinity and tissues. Overall, they decreased, then recovered after alkalinity stress, and increased after infection in the liver and gill, and did not recover after alkalinity stress, but increased after infection in the spleen. These results showed that alkalinity stress and infection with bacteria had different effects on nonspecific immune function.

Toll-like receptor (*tlr*) is one of the key components of innate immunity. Among various TLRs, *tlr5* is involved in recognizing bacterial flagellin and inducing the production of proinflammatory cytokines (Basu et al., 2012).  $\gamma$ -Interferon ( $\gamma$ -*ifn*) is one of the key cytokines involved in the immune response of helper T Lymphocyte 1 (Th1) (Boehm et al., 1997), which helps to improve the antibacterial activity of macrophages such as those in rainbow trout (*Oncorhynchus mykiss*) (Zou et al., 2005) and goldfish (*Carassius auratus* L.) (Grayfer et al., 2010), and strengthen antigen presentation and leukocyte transport (Zou et al., 2005). In this study, it was found that the gene expression level changed more dramatically than the enzyme activity. At 31 d of alkalinity stress and 7 d post-infection, the *tlr5* mRNA expression level in the liver and spleen of the 10 mmol/L AW group was significantly higher than that of the FW and 30 mmol/L AW groups. At 31 d of alkalinity stress,  $\gamma$ -*ifn* mRNA expression level in the liver and gill of the 30 mmol/L AW group was lower than in the initial group. At 7 d post-infection, the  $\gamma$ -*ifn* mRNA expression level in the spleen and gills of the 10 mmol/L AW group was close to that of the FW group, and significantly higher than that of the 30 mmol/L AW group. Thus, it was suggested that high alkalinity would inhibit *tlr5* and  $\gamma$ -*ifn* expression, thereby reducing immune function in tilapia. Interleukin-1 (*il-1*) is a cytokine that plays a key role in the local and systemic inflammatory response. *il-1 $\beta$*  can participate in the expression of immune-related genes, lymphocyte activation, phagocytosis and sterilization (Türker, 2012). It is the first proinflammatory factor identified in fish and the earliest expressed proinflammatory factor in vivo (Zhu et al., 2013). In this study, the *il-1 $\beta$*  mRNA expression in the spleen and gill of the 30 mmol/L AW group was significantly higher than that of the 10 mmol/L AW and FW groups at all time points except in the initial group, which may be due to tissue damage caused by high alkalinity and aggravation of the inflammatory reaction.

The liver, as the metabolic part of the three major nutrients in fish, has a detoxification effect in teleosts (Freitas-Lopes et al., 2017). The liver also contains many immune cells, which can secrete cytokines to complement the defence against invasion by pathogens (Freitas-Lopes et al., 2017). In this study, different from in the spleen or gill, the activities of Lys and Alp in the liver at 31 d of alkalinity stress and after infection were higher than those at the initial stage, indicating the role of Lys and Alp in the liver. Spleen is a secondary immune organ in fish, and the main location for immune cells, including B cells, T cells and macrophages to mature and function. It plays a role in the immune response against pathogen infection (Swirski et al., 2009). In this study, at 7 d post-infection, the activities or expression of Lys, Alp, C3, IgM, *thr5* and  $\gamma$ -*ifn* in the spleen of the 30 mmol/L AW group are lower than those of the 10 mmol/L AW and FW group, suggesting that immune function in the spleen was inhibited upon high alkalinity stress. The gill is usually the first organ of fish to respond to the external environment condition. It is often used as a biomarker of aquatic toxicology (Gunal et al., 2008). It has immune tissues with a large mucosal area and strong plasticity. It plays an important role in gas exchange and maintaining electrolyte and acid-base balance (Sollid and Nilsson, 2006). In this study, at 1 d of alkalinity stress, the activities or expression of Lys, Alp, C3, IgM and *il-1 $\beta$*  in the gills of the 30 mmol/L AW group were higher than those in the 10

mmol/L AW group. It is speculated that due to the direct contact of gills with water, high alkalinity stress will lead to a stronger immune stress response in the gill in the short term. Thus, the experimental results showed obvious tissue differences.

In conclusion, long-term low alkalinity (10 mmol/L) water (by addition of sodium bicarbonate) can improve the immune function of tilapia against *Aeromonas hydrophila*, ultimately reducing the mortality of infected tilapia. Although long-term high alkalinity (30 mmol/L) can increase some immunological indexes, it did not increase resistance to pathogens. Immune function in the spleen of tilapia was inhibited upon high alkalinity stress.

## Statements and declarations

The authors declare no competing financial interests to this work submitted for publication.

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

All data generated or analyzed during this study are included in this published article.

## Ethical statement (Compliance with ethical standard)

I certify that this manuscript is original and has not been published elsewhere in any form or language and will not be submitted elsewhere for publication while being considered by Aquaculture. If accepted, it will not be published elsewhere in the same form, in English or in any other language, including electronically without the written consent of the copyright holder. And the study is not split up into several parts to increase the quantity of submissions and submitted to various journals or to one journal over time. No data have been fabricated or manipulated (including images) to support our conclusions. No data, text, or theories by others are presented as if they were our own. Proper acknowledgements to other works must be given.

The submission has been received explicitly from all co-authors. And authors whose names appear on the submission have contributed sufficiently to the scientific work and therefore share collective responsibility and accountability for the results.

This article does not contain any studies with human participants performed by any of the authors.

Informed consent was obtained from all individual participants included in the study.

This study was approved by the animal ethics committee of Shanghai Ocean University.

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## CRediT authorship contribution statement

**Yan Zhao:** Conceptualization, Methodology, Writing – original draft. **Yanling Wang:** Investigation, Data curation, Software. **Lingyuan Song:** Formal analysis. **Shuaishuai Li:** Validation. **Xiaowu Chen:** Visualization. **Shoujie Tang:** Resources. **Jinliang Zhao:** Supervision, Funding acquisition.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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